

Occupational Lung Diseases

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Outline

- Occupational exposure and pathogenesis
- Toxic inhalation injury
- Occupational asthma
- Byssinosis
- Extrinsic allergic alveolitis
- Inhalation fever
- Berylliosis
- Occupational lung cancer

Occupational exposure and pathogenesis

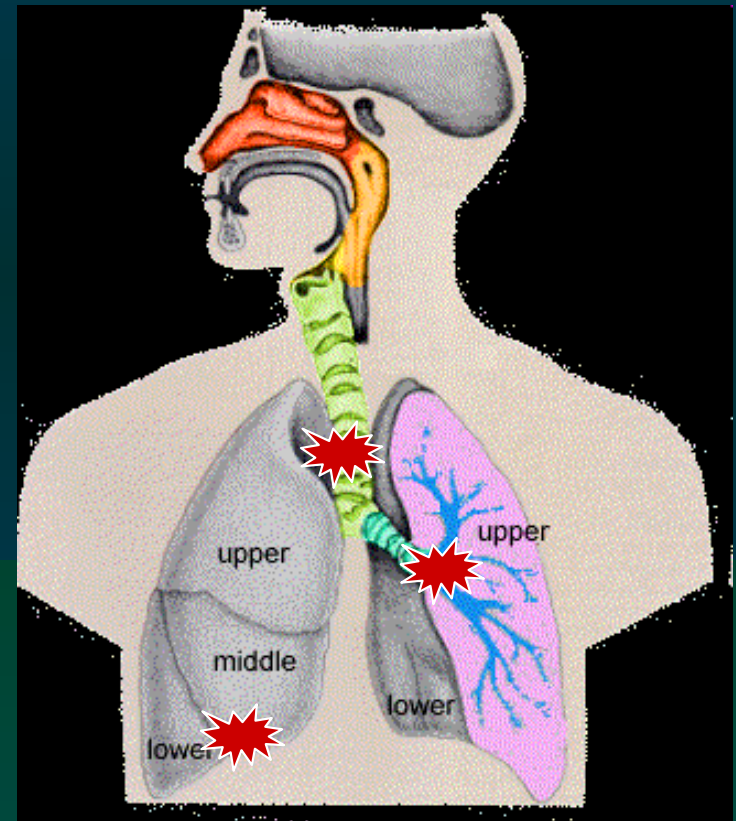
Response of respiratory tract to extrinsic stimuli

■ Acute responses :

1. Nasal sinusitis
2. Pharyngitis
3. Obstruction of airway
4. Alveolitis
5. Pulmonary edema

■ Chronic response :

1. Asthma 氣喘
2. Bronchitis
3. Broncheolitis
4. Lung fibrosis
5. Pleura fibrosis
6. cancer



www.le.ac.uk/pa/teach/va/anatomy/case2/2_1.html

Evaluation of occupational lung diseases

- History of environmental and occupational exposure
- Physical examination
- Adequate study of medical image
- Pulmonary function test

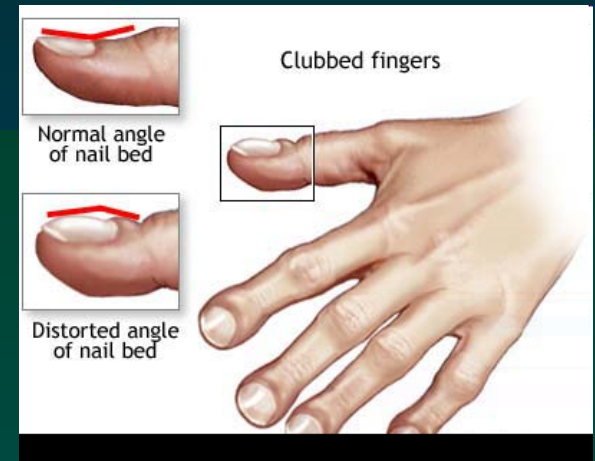
How to collect history of exposure?

- Statements of patient
- Type and time (duration) of exposure
- Environmental control or equipment 通風換氣設備
- Protective devices 個人呼吸防護具
- Material safety data sheet (MSDS) 物質安全資料表
- Field sampling
- Exclude possible exposure in the daily life

Physical examination

- Nonspecific patterns
- cyanosis ? Clubbing fingers ?
- Skin and eyes ? --» inflammation or irritant sensation ?
- Respiratory sounds
- Cardiovascular system

Cardiovascular



Studies of medical image

- Chest X-ray
- Chest CT scan(10mm):
pleura, mediastinum, hila
- High resolution chest CT scan(1~2mm):
diffuse lung process

Parameters of pulmonary function

- FEV₁ ⇒ 深呼吸後大力呼氣 1 sec 內吐氣的量
- FVC : difference between “obstructive” or “restrictive” lung disease ⇒ 肺活量
- FEV₁ / FVC : 比值 越低越差
- Peak expiratory flow rate(PEFR): self-monitor of airway obstruction ⇒ 簡易型肺功能檢測
- DLco : non-specific index
- Bronchoprovocation test : early, late or dual response?
⇒ 測專一性物質是否導致氣喘

Toxic inhalation injury

毒物吸入導致肺症

Toxic mechanisms

- Exposure to high doses of toxin (gas, fume, droplets) in a very short period
- Usual occur in traffic accident
- Location of injury: decided by physical or chemical characteristic of toxin
- **Toxic mechanism of gas** :
 1. Deposit site: related to water solubility
 2. Duration of exposure
 3. minute ventilation

Solubility 水溶性	Toxic materials	Location of injury(deposition)
high	NH ₃ , formaldehyde	Upper airway
moderate	甲醛 Chlorine, sulfur dioxide	Lower airway
low	氮氧化物 Nitrogen oxides, phosgene	Lung parenchyma
aerodynamic 顆粒大小 (micro-meter)		
> 10	Dust from earth's crust	Upper airway
2.5 ~ 6	Fire smoke particles	Lower airway
< 2.5	Metal fumes, asbestos fibers	Lung parenchyma

Responses of toxin inhalation

■ Severity $\overleftarrow{?}$ $\overrightarrow{?}$ total amounts of exposure

Clinical manifestations

Location	Acute response	Chronic response
eye , nose and sinus , oropharynx	Focal irritant or inflammatory process	Corneal scar , nasal polyp
Upper airway	Laryngeal edema , airway stenosis	Laryngeal polyp
Lower airway	bronchitis	asthma
Lung parenchyma	pneumonia , pulmonary edema , respiratory distress	Lung fibrosis , obstructive broncheolitis

How does total amount of exposure affect clinical manifestations?

■ High solubility(NH₃ or Chlorine)exposure :

- p **Low amount** : irritant sensation of conjunctiva or upper airway
- p **Moderate amount** : hoarseness , cough , bronchospasm
- p **High amount** : acute respiratory distress

劑量

Physical examination

- Burn injury of nasal cavity/hoarseness/wheezing ?
- Chest X-ray :
 1. most of cases are normal in the initial evaluation
 2. suggest observation over 24 hours
 3. acute respiratory distress can occur in following 4-8 hours
- Artery blood gas survey :
early detect hypoxia

Management

- Wash possibly residual toxins on skin or cornea
- Laryngoscopy for early detection of airway stenosis (especially for persons with burn injury on the face) □ → early intubation for airway support
- Early detect stenosis of airway by pulmonary function
- Oxygen and bronchodilator
- respirator

Occupational asthma

職業性氣喘

Nearly 1/5 of adult asthma was related to occupational exposure in USA

What is asthma ?

- Reversible airway stenosis and inflammation
- Airway hyperresponsiveness
- Occupational asthma :
exposure at the work site

Possible pathogenesis

- Type 1 immune response
- Drug effect
- Direct irritant effect on the airway

Possible occupational exposure materials

引發氣喘物質

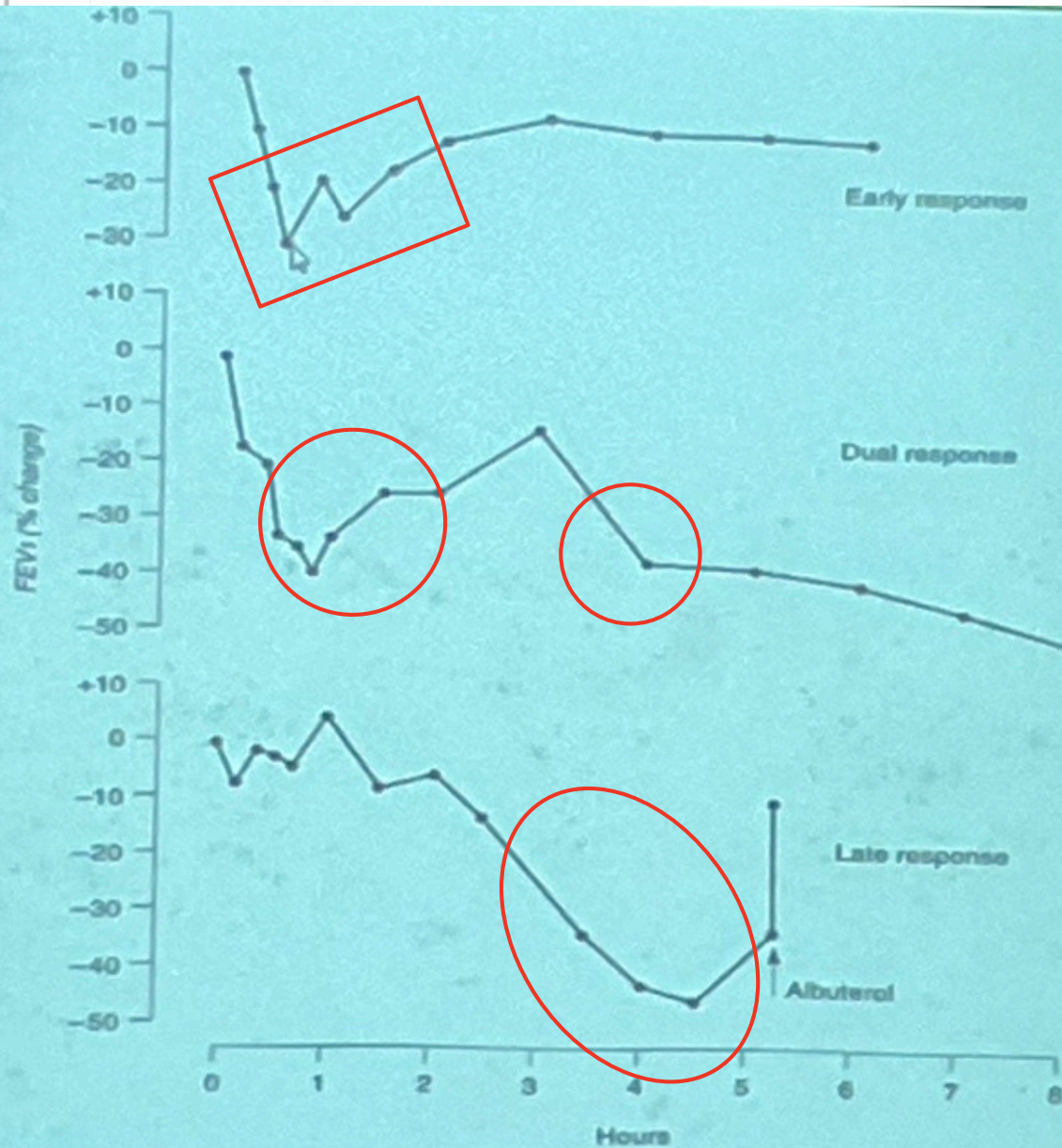
- At least 250 materials were noted to be related to occupational asthma

Pathogenesis

- Eosinophil infiltration
- Airway edema
- Epithelial defect of airway
- hypersensitization : interaction between allergens and macrophages, mast cells?



+10



早發型

雙發型
又發型

遲發型

Figure 20-1. Potential responses to inhalation of allergen in sensitized workers with asthma.

Joseph LaDou : current occupational and environmental medicine



Diagnosis criteria

- Confirm diagnosis of asthma
- **Relation between asthma and occupational environment :**
 1. Asthma attack at work hours or 3-4 hours after work
 2. Recurrence at work days ⇒ 上班日同發 ↳ 下班 3-4 hr
 3. Symptoms improved on vacation days ⇒ 放假就好
 4. Exacerbation at prolonged work hours ⇒ 工時↑嚴重↑
 5. Symptoms subsided after adjustment of work contents ⇒ 換工作就好

Investigate environmental status

- Type of exposure materials °
- Ventilation devices at the work field
- Respiratory protective devices °
- Detect amounts of exposure

Tool of diagnosis

■ Pulmonary function :

best diagnostic tool(FVC,FEV₁)

1.**Bhronchodilator test** : **improve** FEV₁ for 12% or more?

→ [?]positive response.

2.decreased FEV₁ for 20% or more.

3.**Periodically evauation of PEF_R**: performed for at least 4 times a day.Any variation exceeding 20% will be treated as positive response!

4.**Bronchial provocation test** : specific v.s. nonspecific (methacholine, histamine)

■ **Skin hypersensitivity test** ⇒ 皮膚貼片

Other notification

- Prolong exposure may cause chronic asthma which make diagnosis become difficult.

Classification of health management

- First class :

no respiratory symptoms , normal chest X ray , normal lung function

- Second class :

subjective respiratory symptoms , normal chest X ray and lung function

- Third class :

subjective respiratory symptoms with abnormal chest X ray and lung function

Byssinosis

棉絮沉着症

Etiology

- Long term exposure of cotton dusts causes cough, respiratory difficulty and wheezing
- Monday feeling, **Monday fever**
- Symptoms could persist for several days or cause persistent cough and respiratory distress
- Normal chest X-ray

Extrinsic allergic alveolitis

外因性過敏性肺泡炎

Pathogenesis

■ acute :

symptoms occurred at 3 hours after exposure [?] cough ,
fever , respiratory distress , aching soreness of muscle

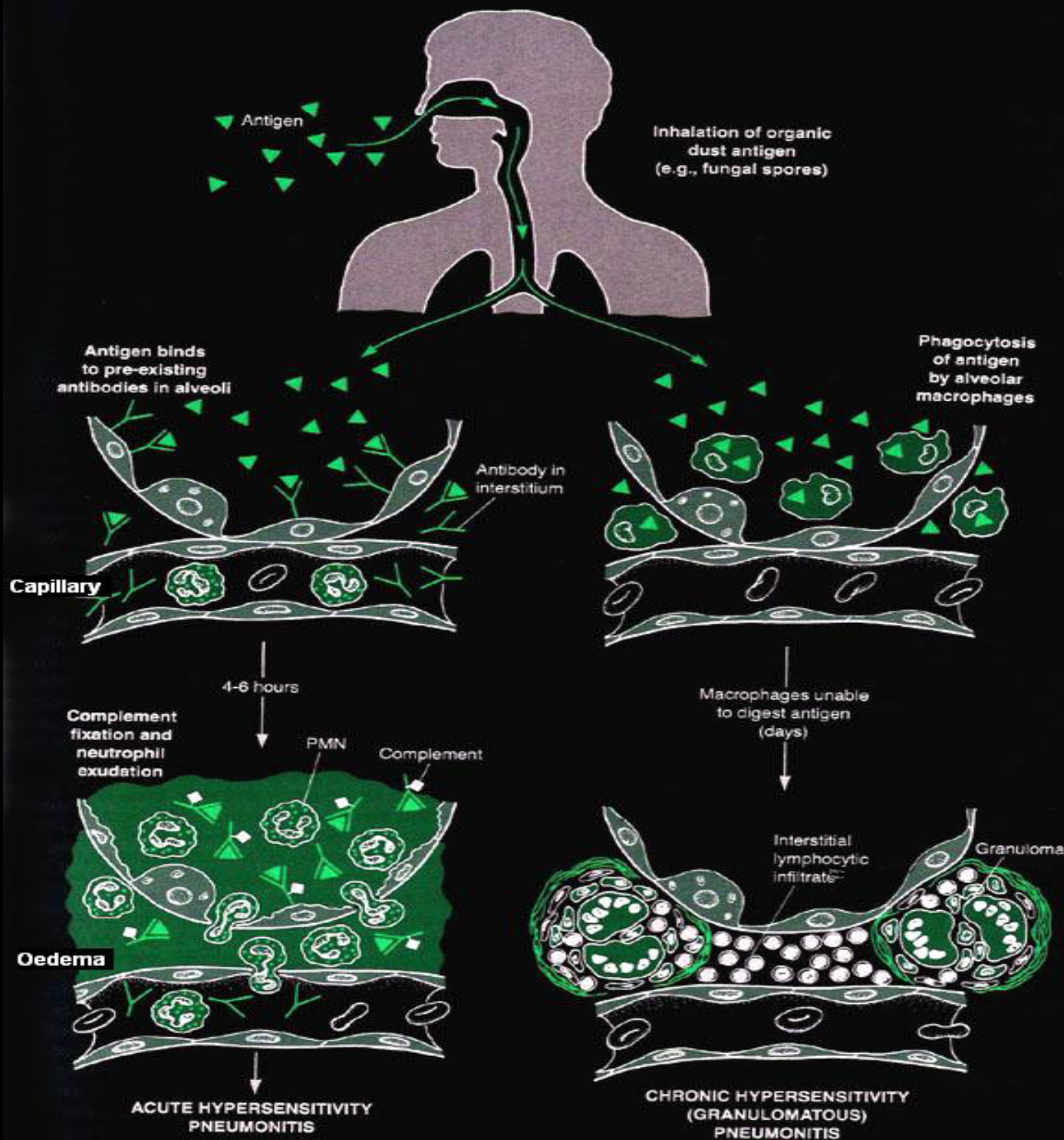
→ [?] disappear 3 days later

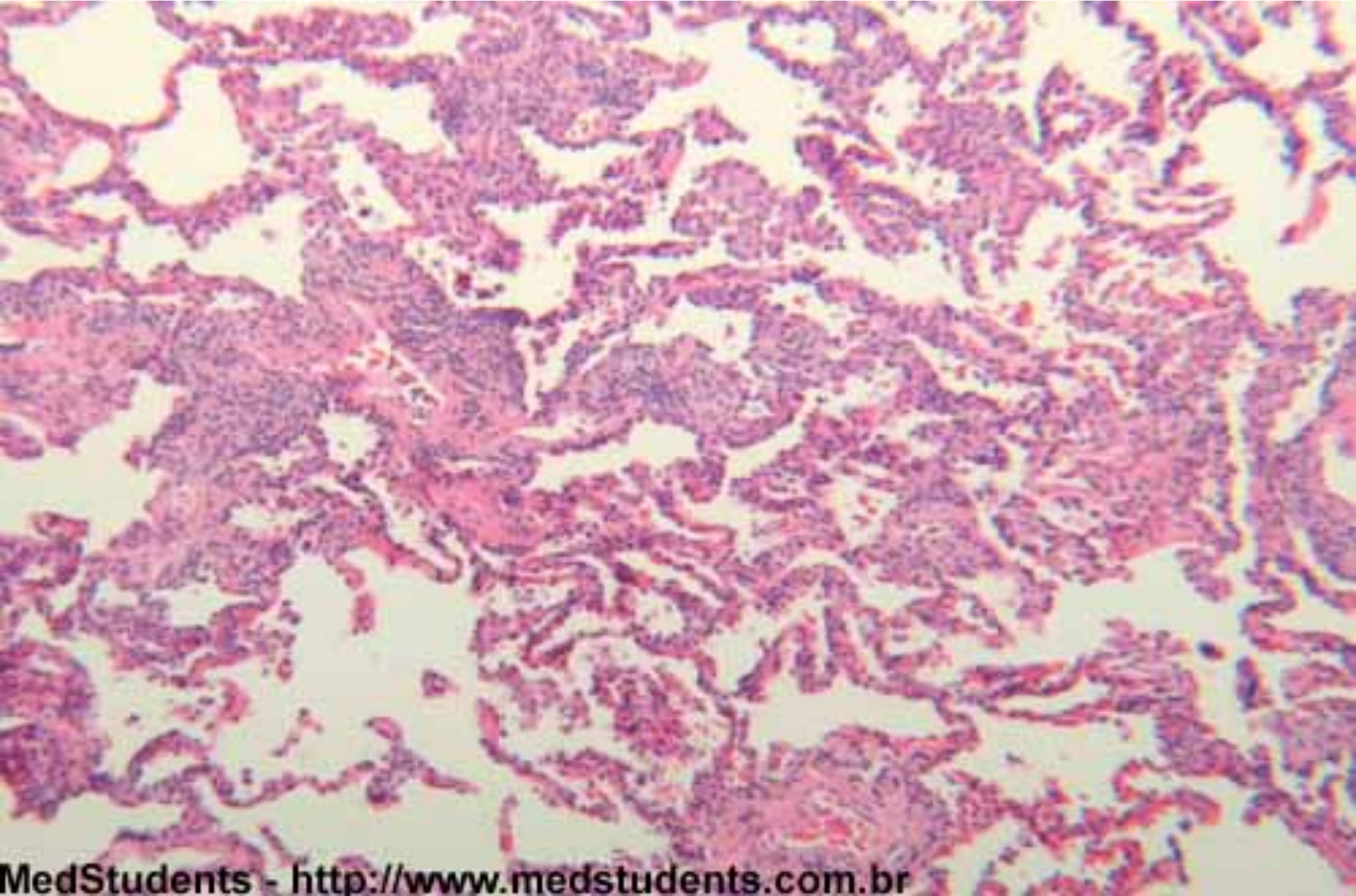
- { 1.nodular or opacity infiltration on
- { 2.biopsy : granuloma in alveolar space or bronchiolar space

後遺症

■ Chronic :

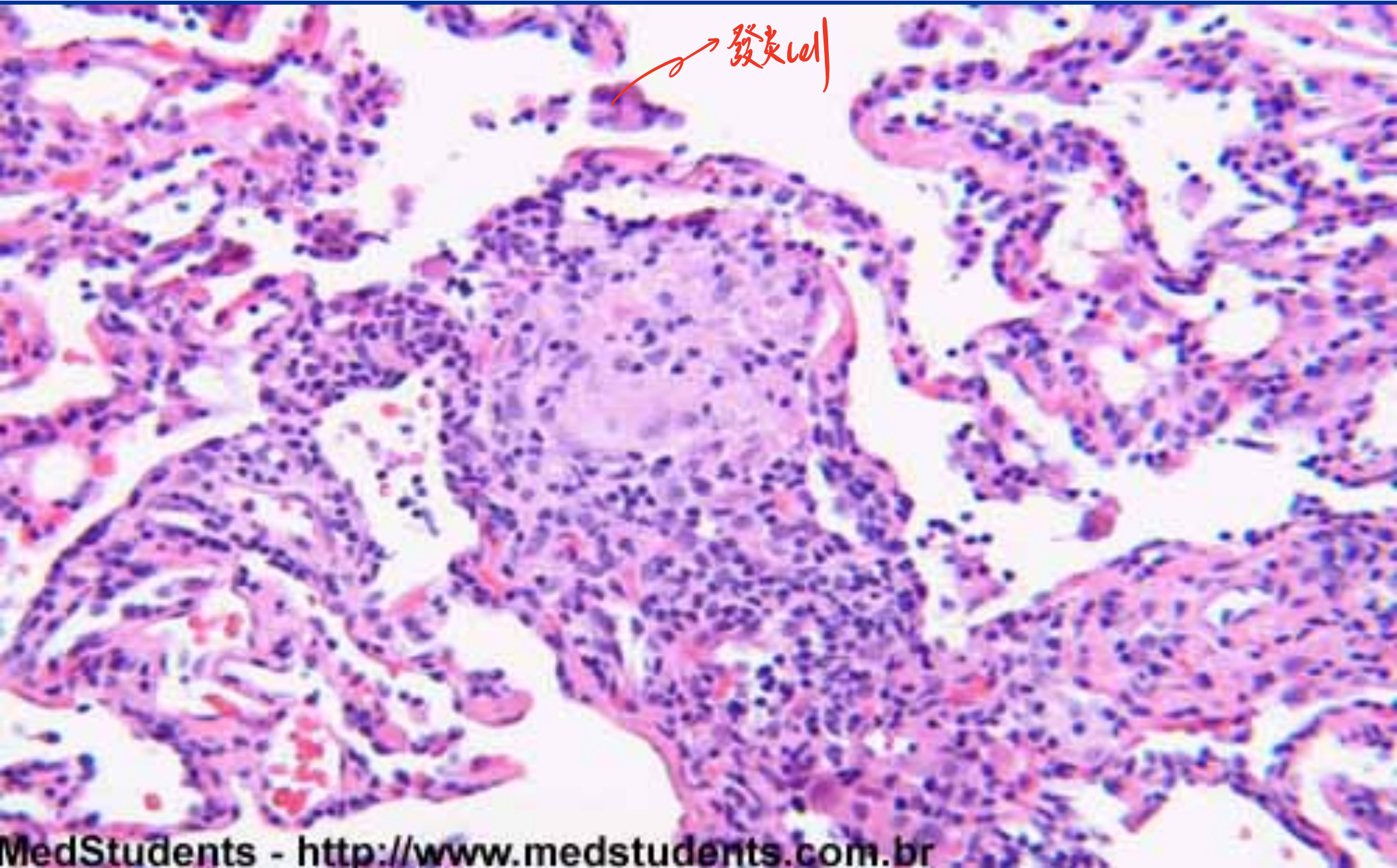
- { lung fibrosis after repeatedly inflammation





MedStudents - <http://www.medstudents.com.br>

Interstitial inflammation and fibrosis
in hypersensitivity pneumonitis (H&E).



Nonnecrotizing granuloma in hypersensitivity pneumonitis (H&E, 160x).

CXR

■ acute :

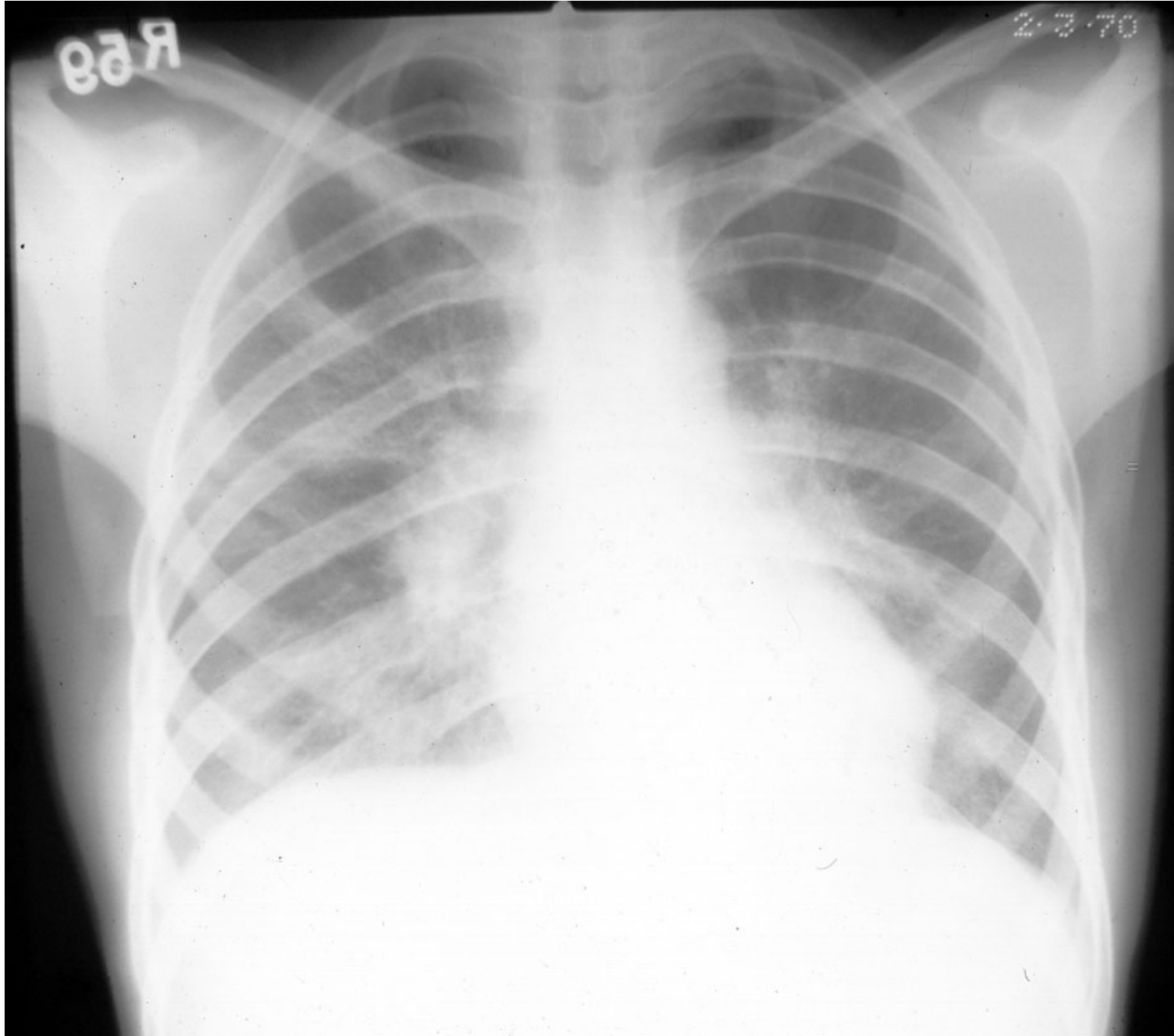
- 1.normal
- 2.reticulo-nodular patterns
- 3.opacities

■ chronic :

- 1.Diffuse linear infiltration
- 2.Interstitial fibrosis

R 53

2-3-70



Diagnosis criteria

■ major :

1. Specific history of exposure ⇒ 暴露史
2. Clinical manifestations , CXR , lung function and physical examination are compatible with interstitial lung disease
3. Rule out other possible causes

■ minor :

1. Specific antibody in the serum
2. Lymphocytosis in biopsy or bronchial-alveolar lavage fluids
3. Recurrence of symptoms after exposure to the same material

Inhalation Fever

吸入性發燒

Pathogenesis

- Symptoms occurred at 3-10 hours after exposure □ sore throat, fever, chills, aching soreness of muscle, fatigue and cough
- CXR , lung function , arterial blood gas :
normal
- Spontaneous remission 1-2 days later

Possible exposures

- Metal fumes : ^{Zn} Zinc , ^{Cu} Copper , Mg
- humidifier fever : contaminated water
- Organic dusts : mold feed , sawdust
- mill fever : cotton , jute
- grain fever : grain dust

Berylliosis

鉍肺症

Diagnosis criteria

- Definite occupational exposure
- Lymphocytosis in blood or bronchio-alveolar lavage
- Granuloma and monocytosis in biopsy
- **Include one of the followings :**
 1. Respiratory symptoms
 2. diffuse infiltration of fibrosis of lung on CXR
 3. Restrictive or obstructive ventilatory impairment ,
decreased DL_{CO}

Occupational lung cancer

Definite carcinogens

- Free radicals
- Asbestoses
- arsenic *As*
- Radon *Ra*
- Dichloro ether
- chromium *Cr*
- Toxic fumes
- mustard gas
- nickle *Ni*
- polyaromatic hydrocarbons

Possible carcinogens

- acrylonitrile
- beryllium
- cadmium
- formaldehyde
- silicon
- Man-made fiber glass
- vinyl chloride monomer

**Thanks for your
attention**